Algebra 9 — Simultaneous equations (one linear, one non-linear) · Proof

Calculator allowed. Marks in brackets. ★ = harder question. Always substitute the linear equation into the non-linear one.

1 Solve the simultaneous equations $y = x^2 - 3x + 1$ and $y = x + 2$, giving your answers to 3 s.f. [4]

Substitute: $x^2 - 3x + 1 = x + 2 \Rightarrow x^2 - 4x - 1 = 0$. Formula: $x = \dfrac{4 \pm \sqrt{16 + 4}}{2} = 2 \pm \sqrt{5}$, so $x = 4.24$ or $x = -0.236$. Then $y = x + 2$: $y = 6.24$ or $y = 1.76$. Solutions $(4.24,\ 6.24)$ and $(-0.236,\ 1.76)$.

2 Solve the simultaneous equations $x - y = 2$ and $x^2 + y^2 = 20$. [4]

From the linear equation $y = x - 2$. Substitute: $x^2 + (x - 2)^2 = 20 \Rightarrow 2x^2 - 4x + 4 = 20 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x - 4)(x + 2) = 0$. $x = 4 \Rightarrow y = 2$; $x = -2 \Rightarrow y = -4$. Solutions $(4, 2)$ and $(-2, -4)$ — the two points where the line meets the circle.

3 Give a counter-example to show that $(x - 3)^2 = x^2 - 9$ is false. [2]

$x = 0$: left side $(-3)^2 = 9$, right side -9. Not equal, so the statement is false. (One counter-example is enough.)

4 Prove that if the difference between two whole numbers is 3, the difference between their squares is a multiple of 3. [2]

Let the numbers be n and $n + 3$. $(n + 3)^2 - n^2 = n^2 + 6n + 9 - n^2 = 6n + 9 = 3(2n + 3)$, which is 3 times a whole number.

5 Prove that the sum of any two consecutive odd numbers is a multiple of 4. [4]

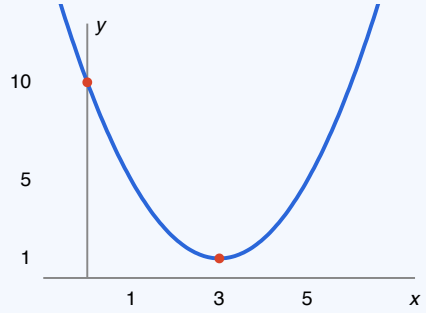
Any odd number can be written $2n + 1$ with n a whole number; the next odd number is $2n + 3$. Sum $= (2n + 1) + (2n + 3) = 4n + 4 = 4(n + 1)$, a multiple of 4. (Writing "$3 + 5 = 8$" is a check, not a proof — the algebra covers every case.)

6 **a** Prove that $x^2 - 6x + 10 > 0$ for all values of x. **b** Find the minimum point of the graph of $y = x^2 - 6x + 10$. **c** Sketch the graph. [9]

a Complete the square: $x^2 - 6x + 10 = (x - 3)^2 - 9 + 10 = (x - 3)^2 + 1$. A square is never negative, so $(x - 3)^2 \geq 0$ and $(x - 3)^2 + 1 \geq 1 > 0$ for every x.

b The smallest value, 1, occurs when $(x - 3)^2 = 0$, i.e. $x = 3$: minimum point $(3, 1)$.

c U-shaped, vertex $(3, 1)$, crosses the y-axis at $(0, 10)$, never touches the x-axis (part a).



Total: 25 marks

Check: substitute the linear into the quadratic/circle, then solve the quadratic — each x gives its own y · a counter-example is a single specific case that fails · proofs use n, $2n$ (even), $2n + 1$ (odd), consecutive = $n, n + 1$ · "always positive": complete the square and argue a square is ≥ 0.